



MTHA6002A History of Mathematics Portfolio

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Table of Contents

Week 2: Group Essay on Archimedes and Reflection	2
Week 3: Group Essay on Ramanujan and Reflection	4
Week 4: Group Essay on Galois and Group Reflection.....	6
Week 5: Reflection on Marked Group Presentation	8
Week 10: Reflection on Marked Individual Presentation	9

Week 2: Group Essay on Archimedes and Reflection

Archimedes' On Spirals

Archimedes was a mathematician in Ancient Greece from the city of Syracuse, Sicily. He lived from 287-212 BC, dying at the age of roughly 75 to a Roman soldier during the Siege of Syracuse despite orders for him to remain unharmed. It's said he was so engrossed in calculations he was unaware the attack was even happening [1]. It is believed that Archimedes' studied in Alexandria before returning to Syracuse where he would spend most of his life doing mathematics and being persuaded to help in war efforts despite his preference for pure mathematics over applications of it.

Archimedes' work was vast, covering war machines, physics and geometry in depth and creating elegant proofs others couldn't see. Despite much of his work being lost to time his work on mathematical physics, however, has survived and includes the law of the lever, foundations of statics, center of mass calculations and Archimedes Principal. He was quoted to have said "Give me a lever long enough and a fulcrum on which to place it, and I shall move the world".

Archimedes was most accomplished at geometry, among the most famous and groundbreaking work is "*On Spirals*". Containing 28 propositions, his "*On Spirals*" work was unique for the time because it moved away from static constructs to focus on curves generated by motion, his work on the subject was extensive [2].

Archimedes first defined what is now well-known as an Archimedean Spiral. He described a line revolving in a plane with a point moving uniformly across it to draw a spiral "If a straight line of which one extremity remains fixed be made to revolve at a uniform rate in a plane until it returns to the position from which it started, and if, at the same time as the straight line revolves, a point move at a uniform rate along the straight line, starting from the fixed extremity, the point will describe a spiral in the plane." [3]. In modern notation this would be $r = a\theta$, with a being the constant velocity of the point and θ the angle. By using this, Archimedes went on to prove many properties of this spiral such as the area bounded by a set number of revolutions. Archimedes then goes on to describe the area created by a spiral in its first revolution as $\frac{1}{3} * A$, where A is the area of the circle with radius equal to the distance between the first and last points [4]. Successive rotations then add linear multiples of the area added by the second rotation, so the area added by the third rotation is $2 * A_2$, with A_2 being the area added by the second rotation, area added by the fourth is $3 * A_2$, fifth is $4 * A_2$ etc [5]. Today we can prove this easily using integration. We define the area of a sector as $\frac{1}{2}r^2 d\theta$ with radius r and angle $d\theta$. Then:

$$A = \int_0^{2\pi} \frac{1}{2} r^2 d\theta = \int_0^{2\pi} \frac{1}{2} (a\theta)^2 d\theta = \frac{1}{3} \pi (2\pi a)^2$$

Comparing this with a circle of the same radius, $r = 2\pi a$:

$$A = \pi r^2 = \pi (2\pi a)^2$$

Therefore, proving Archimedes was correct despite lacking the tools we have today.

References

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Reflection

Upon writing the group essay, my biggest struggle was getting started. I initially struggled to find academic sources due to the age of the topic, which showed me the importance of evaluating academic sources early to write around, which I will attempt to prioritise next time. Producing a first draft made refinement and rewording much easier, allowing me to streamline my writing to a word limit. In hindsight, I found that leaving referencing until the end was inefficient and confusing, meaning points weren't clearly linked to sources. Although initial planning with my group was effective, limited communication and coordination reduced efficiency of our planned method of combining our work. This showed me the importance of establishing clear roles and maintaining communication to avoid having to restructure and improvise later. My takeaway from this is that structure and sources are the most important starting steps for my method of essay writing.

Week 3: Group Essay on Ramanujan and Reflection

Ramanujan's Life and Work on the Partition Function

Ramanujan was an Indian mathematician born in 1887 in Erode, India and died only at the age of 32 in 1920 [1]. Despite his lack of formal training, his discoveries and approximations stunned mathematicians of the time since most of the time he did not have proof, yet his theorems worked [2].

As a child, Ramanujan grew up relatively poor and didn't have access to many resources such as textbooks and mathematical literature that Europeans at the time would have. He learned from two college students who were lodgers at his home but exhausted their knowledge by the age of 11 [3]. He was lent a book on advanced trigonometry at 13 and mastered it whilst discovering theorems on his own [3].

Despite his obvious genius, when Ramanujan reached out to mathematicians he was overlooked due to his unusual ways of presenting, lack of conventional proofs and the thought that intelligence was mostly concentrated in Europe [3], so Ramanujan being self-taught and from India gave him an unfortunate disadvantage at being taken seriously.

In 1913, Ramanujan attempted to get into contact with Godfrey Hardy, a professor at Cambridge and one of Britain's leading mathematical elites. This was because Hardy specialized in number theory and Ramanujan decided he was one of the only people who may understand his work [3]. Despite initial doubts, upon reading his work Hardy was in awe, recognizing it as a product of extraordinary genius, being quoted as saying "They must be true, because if they were not true, no one would have had the imagination to invent them." [1]. Hardy acted immediately and wrote back, arranging financial support and working to bring Ramanujan to Cambridge [1].

During his time in Cambridge, Hardy worked alongside Ramanujan to produce work that is now central to modern mathematics such as his partition function, describing how many ways you could write a number as a sum of positive integers e.g. $p(4) = 5$, since $4 = 4 = 3+1 = 2+2 = 2+1+1 = 1+1+1+1$. Eventually getting an accurate asymptotic formula as follows using analysis:

$$p(n) \sim \frac{1}{4n\sqrt{3}} e^{\pi\sqrt{\frac{2n}{3}}} \text{ as } n \rightarrow \infty,$$

They ended up finding many properties such as divisibility by 5 when the integer ends in 4 or 9 and showing $p(n)$ grows extremely fast using this explicit asymptotic formula [4]. Ramanujan and Hardy also obtained an asymptotic series that had the property that if you stop after x terms, the size of the error is the same order as the $x+1$ term. This means that since $p(n)$ grows so fast for large n , you need a surprisingly small number of terms to compute it accurately. For example, for $p(200)$, Ramanujan and Hardy could prove after 5 correction terms that the error is less than 0.5, so rounding the partial sum must equal the exact integer value of $p(200)$ which is 3,972,999,029,388 [4].

When Ramanujan was in hospital, Hardy visited and had remarked his taxi number 1729 was a dull one and he hoped it wasn't an unfavourable omen. Ramanujan replied, "it is a very interesting number, it is the smallest number expressible as the sum of two cubes in two different ways" [1], showing his incredibly fast intuition with numbers.

Despite moving to Cambridge and publishing multiple papers, most of his work, credited as being decades ahead of contemporary work, wasn't appreciated until after his early death, with Hardy himself admitting that he didn't fully grasp the significance of everything Ramanujan had written [1].

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Reflection

For this week we discovered our roles better and worked more efficiently as a result. Segmenting and structuring the essay in advance improved our clarity on the subject and the presentation organisation. This translated well to the powerpoint, since the members of the group responsible for the slides can have a clear set out way of ordering and labelling the slides. Giving the presentation went okay, I took on too much without practice and seeing the slides last minute but performed well and gave more context than the points on the slides due to my research and knowledge on the subject. I presented less detail of the partition function than expected since I knew about a series version which had amazing accuracy, however this wasn't a point on the slide so the audience wasn't confused by its absence, emphasising the need to balance depth of the topic with audience clarity.

Week 4: Group Essay on Galois and Group Reflection

Galois and His Work on Polynomials

Évariste Galois was born 25th October 1811 in Bourg-la-Reine in the French Empire. His father was headmaster of a local school; however, when Galois was four, his father became Mayor, laying the foundations for Galois' future political activism [1].

Galois' mother was highly educated and homeschooled him in Latin, classical literature, and religious skepticism. His love for mathematics developed after enrolling at Lycée Louis-le-Grand, Paris, in 1823 [1]. Studying the works of mathematicians like Joseph-Louis Lagrange, he quickly excelled, surpassing his classmates and the curriculum. At 16, he applied to the prestigious École Polytechnique, failing the oral exam. At 18, he tried and failed the oral exam again, arguing with professors and throwing an eraser at one. Following this, Galois was accepted into École Normale, a slightly less prestigious institute [1].

During this time, there was growing unrest towards King Charles X. Around Galois' second rejection, his father fell victim to political attacks before committing suicide [1]. During the 1830 July revolution, Galois was enrolled in university but was keen to take part in the overthrow of Charles X; however, the students were reportedly locked inside to prevent them from joining, the criticism of which later led to Galois' expulsion in 1831 [2]. After Charles came Louis-Philippe I, but Galois still wasn't satisfied. Soon after, Galois joined the Société des Amis du Peuple, becoming increasingly politically active.

In 1831, he was arrested after making a toast interpreted as a threat to the monarch; he was later acquitted but arrested again during a political demonstration, spending months in prison [2]. However, this time in prison was not wasted. Galois began developing original mathematical ideas at 16, getting his first paper published at 17. During his time at university, he began his most known work on the Galois Theory, refining this work while in prison [2].

More of his most known work was on quintics and why there is no formula to solve a 5th degree polynomial. The Ruffini–Abel Theorem was published in 1799, proving there is no general formula using radicals to solve all quintics; however, it didn't explain why and didn't delve into group theory. Galois' addition to this problem was to introduce Galois groups. This associated a polynomial to a group of permutations of its roots, to prove a polynomial is solvable if and only if its Galois group is solvable i.e. it has subgroups whose successive quotients are abelian, changing this quintic problem into a structural, group theory problem. For a general quintic, let S_5 be the Galois (symmetric) group containing the alternating group A_5 . Since A_5 is a non-abelian simple group, S_5 cannot have a solvable series, meaning S_5 is not solvable, and therefore the quintic is not solvable by radicals, as its Galois group S_5 is not solvable. He also showed that for specific cases, his theory holds true. For example, for $x^5-1=0$ the equation becomes solvable as its Galois group is cyclic, so abelian, and therefore solvable. [3]

Galois' death came soon after his release from prison in 1832. Following hurriedly written letters summarising the foundations of his Galois Theory, he participated in a duel, possibly over a romantic conflict, receiving a gunshot to the abdomen, succumbing to his wounds at 20 years old [1].

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Group Reflection

As a group, we found the task this week easier than the last since we have established roles and understand what each member is supposed to be doing. The outcome of this has been increased workflow due to different roles bouncing off one another. The role to set up the files and get a general format forward has helped get the PowerPoint started even if the essay hasn't been written yet and build together a structure that anyone can contribute to, increasing efficiency and quality of work. The role of finding and reading sources has also been very helpful for providing everyone with sources to read and understand before creating the essay or PowerPoint and keeps everyone on the same page.

Week 5: Reflection on Marked Group Presentation

Reflection

The group presentation topic choice could have been a lot better, the choice made by our group consisted of a topic that was not above our current mathematical knowledge since mathematical induction was a first year, if not a level, topic. Originally the group did not decide to make the presentation historical until I had to remind them the module title and expressed my concern for the topic we chose. Reflecting on this I could have made my objections clearer and explained my case better instead of letting the presentation continue and receive the feedback I had expressed concern about in the first place.

Week 10: Reflection on Marked Individual Presentation

Reflection

In comparison to presenting as a group I found the individual presentation more effective for my working style, in the topic choice, presentation creating and giving the presentation itself. My experience presenting was much less stressful individually, I found this because I didn't have to wait and rely on replies from groupmates over who is doing what job, who is presenting etc, which allowed me to build the full presentation myself early so that I had the full knowledge to talk around slides and had practice giving the presentation so that I could get through the slides efficiently and get across ideas without the nervousness of forgetting what to say or not knowing the topic in its entirety.